



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Metody uczenia reprezentacji dla generalizacji modeli świata i polityk z symulacji do rzeczywistości w fizycznej sztucznej inteligencji / Representation learning methods for the simulation-to-reality generalization of world models and policies in physical AI

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	Doctoral School courses. Reading the literature and considering possible research directions
Semestr 2	Doctoral School courses and literature review. Choosing the dissertation topic with the supervisor and preparing the Individual Research Plan
Semestr 3	Setting up world models, BridgeData V2 and SIMPLER on the WCSS supercomputers. First experiments on RQ1 at the K46 department. Building the pipeline for simulation data and its photorealistic transfer (RQ2)
Semestr 4	Completing the RQ1 study in manipulation and analysing its results (H1). Preparing and submitting the first article to a 200-point venue such as ICLR or IEEE RA-L. First experiments on RQ2
Semestr 5	Completing the RQ2 study and analysing its results (H2). First experiments on RQ3. Preparing a second publication
Semestr 6	Completing the RQ3 study (H3). Experiments on RQ4 in unseen scenes and in navigation. Tests on a real robot in a university laboratory if access allows
Semestr 7	Completing the RQ4 study and analysing all results (H4). Submitting a paper to a conference or journal. First draft of the dissertation
Semestr 8	Completing, revising and submitting the doctoral dissertation



4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Robots that learn by trial and error need far more attempts than a real robot can make safely or cheaply, so they are usually trained in simulation. Hand-built physics simulators are fast, but a realistic copy of each new place takes a lot of work, and their images still look different from real camera images. A newer option is a learned world model [1]. It is a neural network that predicts what the robot will see after it takes an action. Recent world models are trained on large amounts of real video and can serve as simulators. UniSim [2] trained robot policies only inside such a model and then used them on real robots, and NVIDIA releases its Cosmos models [3] as a base for building such simulators.

This gives a loop from reality to simulation and back. Real recordings train a world model, a policy is trained inside it, and the policy is then used in the real world. The weak point is that an imagined rollout is never exact. The model may ignore an action or predict motion that breaks physics [4],[5], so a policy that succeeds in the world model can fail on a real robot. Better-looking video does not fix this, because the visual quality of a world model does not reliably predict whether agents that plan with it succeed at their tasks [5]. Physics simulation data are cheap and unlimited, yet in the first such study I have found, a world action model trained only on simulated demonstrations succeeded in about a third of the trials on a real robot arm [6]. I have found no controlled comparison of ways to build a world model from such data so that it predicts the real world well, and no settled answer to how a world model or a policy should be adapted to a new real place with few real samples.

I want to approach these problems through the representations the models learn. A neural network first turns an image into internal features and then decides from them. How these features are learned is a choice: what the world model predicts, which data and encoder it starts from, and which objectives make the features ignore what differs between simulated and real images. LeCun argued that a world model should predict in such a feature space rather than in pixels, so that it can ignore details that cannot be predicted and do not matter for control [7]. My research aims to find out which of these choices let a policy trained in a world model work on real data, how to build a world model from simulation data so that it predicts the real world, and how to prepare both models so that a new real place needs little real data. Manipulation will be my main task, and I will check the main results in unseen scenes and in robot navigation.

The results could let a robot learn a new job in a warehouse, a hospital or a home mostly from simulated and predicted experience and then work there after a short real adaptation. The same



questions arise in autonomous driving, where generative world models already produce scenarios for training and testing, and in World Action Models, which predict the future and choose actions in one network.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The idea of training a controller inside a learned model of the environment goes back at least to Ha and Schmidhuber [1], whose agent learned to play a game inside its own "dream" and then played the real game. Since 2023, large generative models trained on internet and robot video have been used as interactive simulators of the real world. Yang et al. [2] showed with UniSim that policies trained purely in such a model transferred to the real world without further training. Bar et al. [8] built Navigation World Models (NWM), which predict the first-person video of a moving robot and plan by simulating candidate trajectories. NVIDIA's Cosmos platform [3] provides open world foundation models that are meant to be post-trained for particular robots and vehicles. NVIDIA also builds physics simulators, Isaac Sim and Isaac Lab [9], and uses Cosmos Transfer [10] to make their renderings look real. Cosmos Transfer generates video from depth or segmentation maps, so a rendering keeps its layout and motion and the simulated actions remain valid. DreamGen [11] fine-tunes a video world model on the target robot, generates videos of new behaviours, labels them with actions inferred from the video and trains a policy on them. In this way a humanoid robot learned 22 new behaviours from teleoperation data of a single pick-and-place task.

A newer line merges the world model and the policy into a World Action Model (WAM), one network that predicts both future states of the world and the robot's actions. DreamZero [12], a well known model of this kind, reported that it generalized to new tasks and environments more than twice as well as vision-language-action models. The transfer of WAMs from simulation to reality has barely been studied. The first such work I have found [6] fine-tuned a Cosmos video model into a policy, trained it only on synthetic demonstrations and reached 35% average success on a real Franka arm without any real robot demonstrations.

A second line asks what a world model should predict. LeCun [7] argues that it should predict the representation of the future observation rather than its pixels, so that it can drop unpredictable detail such as texture or leaves moving in the wind. DINO-WM [13] predicts the frozen features of DINOv2, a vision network pretrained without labels, and plans without reconstructing pixels. Assran et al. [14] pretrained V-JEPA 2 on more than a million hours of video, and its action-conditioned version, further trained on less than 62 hours of robot video, planned pick-and-place on real Franka arms in labs it had not seen. SkyJEPA [15] trained such a world model on automatically generated data and reported zero-shot simulation-to-reality control of a quadrotor in outdoor flights, with state inputs rather than images. In a study of world models for vision-based quadrotor navigation, the model that scored best in simulation failed on the real platform, while the robustness of the learned representation across environments predicted real transfer [16]. Nilaksh et al. [17] compared six encoders as the base of a world model trained on BridgeData V2, a real manipulation dataset, and found that encoders pretrained on semantic content were generally better for planning and for evaluating a fixed policy than encoders trained to reconstruct pixels. In navigation, RAE-NWM [18] predicted DINOv2 features instead of the autoencoder codes of NWM and planned better with them on SACSoN and SCAND but not on RECON. Neither work trained policies inside the models, and I have found no controlled



comparison of how the prediction target and the pretrained encoder affect the performance on real data of policies trained inside a world model.

World models are also used to evaluate and improve policies. Ctrl-World [19], trained on the DROID dataset, ranked policies without real rollouts and improved a policy by fine-tuning it on imagined successful trajectories. World-in-World [5] tested world models in closed loop and found that visual quality does not guarantee task success, while how well a model follows actions matters more. The authors of VLAW [4] found that current world models lack the physical accuracy needed to improve policies. For physics simulators, Li et al. [20] showed with SIMPLER that scenes visually matched to real images can rank policies in nearly the same order as real runs.

Almost all large world models are trained on real video. Training them on physics simulation data is attractive, because such data are cheap, come with exact actions and can be generated without limit, but it moves the simulation-to-reality gap into the world model. The classic tools for this gap were built for policies. Domain randomization [21] varies simulator parameters such as textures so that reality looks like one more variation, and co-training on simulated data together with a small real set improved real-world manipulation [22]. For world models, SimDist [23] pretrained a world model in simulation and adapted only the part that predicts the next state with real data. Wang et al. [24] fine-tuned the encoder and policy of a world model pretrained in simulation on a small set of real demonstrations, with object poses rather than images as input. I have found no controlled study of which representation learning choices, such as appearance randomization or transfer, an objective that pairs simulated and photorealistic views, or co-training, make a world model trained on simulation data predict real recordings well.

When a small amount of real data is available, it is used in different ways. VLAW [4] fine-tunes the world model on real rollouts and then improves the policy with synthetic data from it, DreamGen [11] fine-tunes its video world model on the target robot, and LoRA [25] adapts large pretrained models cheaply by updating only a small set of added weights. SimDist [23] trained its encoder on simulation data and kept it frozen during real adaptation. I have not found a comparison of how the representation of a world model, and of the policy trained in it, should be pretrained so that the fewest real samples are needed later. None of these works asks how the two models should learn their representations so that a policy trained in a world model works on real data after little real adaptation, which is the gap my research questions in section 7 address.

The works cited in this document are listed in section 12.

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The goal of my dissertation is to develop representation learning methods that let learned world models serve as simulators for robot learning that generalizes to the real world. By representation learning I mean how the world model and the policy learn their internal features: what the world model predicts, which data and encoder they start from, and which objectives make the features ignore what differs between simulated and real images. I will judge these choices only by observable results.

My work has to fit several constraints. I will fine-tune existing pretrained world models and policies on academic GPU clusters instead of training large models from scratch. I have no robot of my own, so I will work on public datasets and test on a real robot only if access allows. By performance on real data I mean how closely a policy matches the actions of held-out real recordings and how often it succeeds in SIMPLER, a simulator of scenes rebuilt from real recordings. This is a proxy for success on a real robot. Manipulation is my main task, and navigation serves as a check.

My research questions and hypotheses are as follows:



1. RQ1. Which representation learning choices affect how well a policy trained in a world model built from real data performs on real data, compared with a policy trained directly on that data?
2. H1. With the same predictor and data, a world model that predicts the compressed features of an encoder pretrained on real video gives better policies than one that predicts the codes of a pixel autoencoder, and the gain remains when both follow actions equally well.
3. RQ2. Which representation learning choices make a world model trained on physics simulation data predict real recordings well and give policies whose actions match recorded real actions?
4. H2. Training the encoder to give the same features for a simulated frame and its photorealistic version helps more than only adding the photorealistic frames to the training data, unless the gap comes mainly from the simulated physics.
5. RQ3. How should the encoder of a world model and a policy be pretrained so that adapting them to new real scenes needs less real data?
6. H3. An encoder further pretrained on real and simulation data with the objective from H2 reaches the performance of fine-tuning on all available real data with less real data than the other pretraining variants.
7. RQ4. Do these answers hold in scenes unseen during training, and does the answer to RQ1 hold in robot navigation?
8. H4. The choices that work best in manipulation also improve performance on real data over the same baseline in unseen scenes and in navigation.
9. Fallback. If the representation choices make no clear difference, I will instead study how the mix, amount and diversity of generated, simulated and real data affect performance on real data.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The main expected contribution of my dissertation is a set of representation learning methods that let learned world models serve as simulators for robot learning that generalizes to the real world. It should give evidence on which choices, such as the prediction target and the pretrained encoder, affect how well policies trained in a world model perform on real data, and whether these choices still help in unseen scenes and in navigation (RQ1, RQ4). It should also give training methods that make world models built from physics simulation data predict real recordings better (RQ2), and pretraining methods that let a world model and a policy adapt to a new real setting with less real data than full fine-tuning (RQ3).

World models are becoming a common way to create training data and to evaluate robot policies, and World Action Models now put the world model and the policy into one network. My results should show which properties of a world model must be right for a given task, which is useful both to those who build world models and to those who train policies in them. The results on pretraining and on objectives that ignore the difference between simulated and real images may also be useful in domain adaptation more broadly. I plan to release the code, the evaluation protocol and, where licences allow, the trained models, so that others can compare methods in the same way.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).



My work will be mostly empirical. I have not found an established theory of how world models, or policies trained in them, generalize to reality, so I will run controlled comparisons in which one choice changes at a time. I will not train large models from scratch. I will fine-tune existing pretrained world models, encoders and policies whose code and weights are public. My main setting is manipulation on BridgeData V2, a public dataset recorded with a WidowX robot arm, together with SIMPLER [20], which rebuilds some BridgeData V2 scenes in a physics simulator. For H1 I will train the same predictor on the same data and swap only what it predicts: the codes of an image autoencoder trained to reconstruct pixels or the features of an encoder pretrained on real video, such as V-JEPA 2 [14]. I will follow the setup of Nilaksh et al. [17], whose code is public. I will train a policy in each world model by generating imagined trajectories once, keeping those that one success classifier judges successful, and fine-tuning the policy on them. For navigation I will use RAE-NWM [18] on the RECON and SCAND datasets, carry over only the choices that won in manipulation and keep the released settings of RAE-NWM.

For RQ2 I will generate simulation data with scripted solutions of the tasks in the SIMPLER scenes and add random variations of their appearance and physics. If this simulator limits the variations I need, I will use NVIDIA Isaac Lab [9] instead. I will make the rendered frames look real with Cosmos Transfer [10], driven by depth images. The world models trained on these data will differ in one choice: appearance randomization, photorealistic transfer, physics randomization, an objective that gives each simulated frame and its photorealistic version the same features, or co-training with a small real set. For RQ3 I will continue pretraining smaller released encoders on real data, or on real and simulation data, and adapt the world model and the policy to new scenes with a few small amounts of real data, by full fine-tuning or with LoRA [25].

I have no robot of my own, so I will measure on public data. For a world model I will measure how well it predicts held-out real recordings, with standard image and video similarity scores such as PSNR, LPIPS and FVD. For a policy I will measure how closely its actions match those of held-out real recordings and how often it succeeds in SIMPLER, where it acts in closed loop. In RQ2 I will not count success in SIMPLER as evidence, because the training data come from the same simulator. I will also measure how faithfully each world model follows actions and compare prediction targets at similar action following, using earlier checkpoints where needed, so that the effect of the target is not confused with it. In navigation I will measure how close the path of the policy is to the recorded one and whether it reaches the goal. For RQ3 I will report performance as a function of the amount of real data, which shows how much real data each variant needs. For RQ4 I will compare the gain of each best choice in the training scenes with its gain in unseen scenes and in navigation. All of these are a proxy for success on a real robot, and I will say so with every result. If access allows, I will check selected results on a real robot in a university laboratory.

To avoid bias, I will leave the BridgeData V2 recordings of the scenes rebuilt in SIMPLER out of world model training and use them only for adaptation and testing. As a baseline I will train the same policy directly on the real data used to build the world model, to see whether imagined data help at all. I will compare my methods with domain randomization [21] and co-training on simulated and real data [22], with the same training budget and the same amount of real data. I will repeat runs with several random seeds, keep the data for training, model selection and testing separate, and check the success classifier by hand on a sample of rollouts from every world model.

I will use PyTorch for training, Hugging Face for downloading and publishing models and data, and Git with GitHub for version control. I will run experiments on the H100 nodes of the Wrocław Centre for Networking and Supercomputing (WCSS) and on PLGrid, with jobs managed by SLURM. I will respect the licences of the open weights, some of which allow only non-commercial use. I intend to use AI coding agents such as Claude Code to write and test code faster and to watch over long experiments.

I will release my code and, where licences allow, trained checkpoints. I plan to publish at conferences and in journals worth 200 points on the ministerial list, such as NeurIPS, ICML, ICLR, CVPR, RSS and IEEE RA-L, and to post the papers as arXiv preprints.



10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Roboty mogą uczyć się nowych umiejętności w komputerowej kopii świata. Jest to bezpieczniejsze i tańsze niż nauka na prawdziwym robocie. Jedną z takich kopii jest model świata. To sieć neuronowa, która obejrzała wiele nagrań wideo i przewiduje, co robot zobaczy po wykonaniu ruchu. Te przewidywania nigdy nie są jednak dokładne. Robot, który nauczył się czegoś w modelu świata, w prawdziwym świecie często sobie nie radzi. Chcę sprawdzić, jak modele świata i roboty powinny uczyć się widzieć świat, aby ich umiejętności działały także w rzeczywistości. Zbadam też, jak budować modele świata z tanich symulacji komputerowych i jak przygotować robota, aby w nowym miejscu wystarczyło mu niewiele prawdziwej praktyki. Zajmę się głównie robotami, które chwytają i przedstawiają przedmioty, a wyniki sprawdzę też na robotach, które jeżdżą. Chcę opracować metody, dzięki którym roboty będą uczyć się nowej pracy szybciej, taniej i bezpieczniej.

Abstract for general public

Robots can learn new skills in a computer copy of the world. This is safer and cheaper than learning on a real robot. One such copy is a world model. It is a neural network that has watched many videos and predicts what a robot will see after it moves. But these predictions are never exact. A robot that learned something in a world model often struggles in the real world. I want to find out how world models and robots should learn to see the world so that their skills also work in reality. I will also study how to build world models from cheap computer simulations and how to prepare a robot so that it needs only a little real practice in a new place. I will mainly work with robots that grasp and move objects, and I will also check the results on robots that drive around. I want to develop methods that let robots learn new work faster, more cheaply and more safely.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in



the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

September 2027 (planned submission to a venue worth 200 points on the ministerial list, such as the ICLR 2028 conference or the journal IEEE Robotics and Automation Letters).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

I used Claude (via Claude Code) for literature search, drafting and editing, and I reviewed all content and references myself.

Bibliography

- [1] Ha, D., & Schmidhuber, J. (2018). World models. arXiv:1803.10122.
- [2] Yang, S., et al. (2024). Learning interactive real-world simulators. ICLR.
- [3] Agarwal, N., et al. (2025). Cosmos world foundation model platform for physical AI. arXiv:2501.03575.
- [4] Guo, Y., et al. (2026). VLAW: Iterative co-improvement of vision-language-action policy and world model. ICML.
- [5] Zhang, J., et al. (2026). World-in-World: World models in a closed-loop world. ICLR.
- [6] Wang, Z., et al. (2026). Efficient sim-to-real transfer of world-action models from synthetic priors. arXiv:2606.31101.
- [7] LeCun, Y. (2022). A path towards autonomous machine intelligence. OpenReview.
- [8] Bar, A., et al. (2025). Navigation world models. CVPR.
- [9] Mittal, M., et al. (2025). Isaac Lab: A GPU-accelerated simulation framework for multi-modal robot learning. arXiv:2511.04831.
- [10] Abu Alhaija, H., et al. (2025). Cosmos-Transfer1: Conditional world generation with adaptive multimodal control. arXiv:2503.14492.
- [11] Jang, J., et al. (2025). DreamGen: Unlocking generalization in robot learning through video world models. CoRL.
- [12] Ye, S., et al. (2026). World action models are zero-shot policies. arXiv:2602.15922.
- [13] Zhou, G., et al. (2025). DINO-WM: World models on pre-trained visual features enable zero-shot planning. ICML.
- [14] Assran, M., et al. (2025). V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. arXiv:2506.09985.
- [15] Rao, P., et al. (2026). SkyJEPA: Learning long-horizon world models for zero-shot sim-to-real control of quadrotors. arXiv:2606.23444.
- [16] Zanatta, L., Malczyk, G., & Alexis, K. (2026). Generalization of world models under environmental variability for vision-based quadrotor navigation. arXiv:2606.05015.
- [17] Nilaksh, Jha, S., Zholus, A., & Chandar, S. (2026). Reconstruction or semantics? What makes a latent space useful for robotic world models. arXiv:2605.06388.
- [18] Zhang, M., et al. (2026). RAE-NWM: Navigation world model in dense visual representation space. arXiv:2603.09241.
- [19] Guo, Y., et al. (2026). Ctrl-World: A controllable generative world model for robot manipulation. ICLR.
- [20] Li, X., et al. (2024). Evaluating real-world robot manipulation policies in simulation. CoRL.
- [21] Tobin, J., et al. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. IROS.
- [22] Maddukuri, A., et al. (2025). Sim-and-real co-training: A simple recipe for vision-based robotic manipulation. RSS.



- [23] Levy, J., et al. (2026). Simulation distillation: Pretraining world models in simulation for rapid real-world adaptation. RSS.
- [24] Wang, Y., et al. (2025). A recipe for efficient sim-to-real transfer in manipulation with online imitation-pretrained world models. arXiv:2510.02538.
- [25] Hu, E. J., et al. (2022). LoRA: Low-rank adaptation of large language models. ICLR.

13. Określenie planowanej formy współpracy z promotorem / **Planned form of cooperation with the supervisor**

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

My supervisor and I plan to meet every week. At each meeting I will report my progress and results, and we will agree on the next steps and discuss our joint papers. I will also take part in the weekly meetings of my supervisor's group of PhD students. Between meetings we will stay in touch by email and instant messaging. We will keep code, trained models and the LaTeX sources of papers in shared Git repositories.

14. Opinia promotora pomocniczego / **Opinion of assistant supervisor**

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / **Signatures of doctoral student and supervisor(s)**

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor